

Lab 12 Worksheet — Cables and Connectors —

Identification and Selection

Name: ____ Date: ____

Exam objective: Identify every cable and connector on the A+ objective list by sight and select the correct one for a stated requirement (Core 1 objectives 3.1 and 3.2).

Record as you go

Step	What you did	What you observed
1	Build the identification table with columns for connector name, cable type, what it carries, maximum speed, maximum distance and typical use.	
2	Identify the network group: RJ45 for Ethernet, RJ11 for telephone, F-type for coaxial, LC, SC and ST for fibre, and record what distinguishes each visually.	
3	Record the twisted-pair categories with their speed and distance limits: Cat 5e at 1 Gbps to 100 m, Cat 6 at 10 Gbps to 55 m, Cat 6a at 10 Gbps to the full 100 m.	
4	Identify the video group: HDMI, DisplayPort, DVI and VGA, and record for each whether it carries digital or analog video and whether it carries audio.	
5	Identify the peripheral group: USB-A, USB-B, USB-C, micro-USB, mini-USB, Lightning and Thunderbolt, and record the speed each generation supports.	
6	Identify the storage group: SATA data and power, eSATA, M.2, and the legacy IDE/PATA 40-pin and SCSI 80-pin ribbon connectors.	
7	Solve scenario one: a 120-metre run between	

Step	What you did	What you observed
	two buildings needs 1 Gbps. Copper cannot exceed 100 m, so fibre is the only correct answer — record your justification.	
8	Solve scenario two: a cable run passes beside fluorescent lighting and a motor room. Record why shielded twisted pair or fibre is required and unshielded is not.	
9	Solve scenario three: a user needs to connect a modern laptop with only USB-C to an old VGA projector. Record why an active adapter is required, since VGA is analog and USB-C is digital.	
10	Solve scenario four: a 4K display at 120 Hz is needed. Record which connector versions support this and which do not.	
11	Solve the remaining scenarios covering an external drive at maximum speed, a multi-monitor daisy chain, a legacy serial console connection and a PoE camera run.	
12	Complete the selection guide organised by the question asked: how far, how fast, digital or analog, and does it carry power.	

Verification

Success criterion: Your identification table covers at least 20 connectors with speed and distance limits; every scenario answer names a specific cable or connector and justifies it against a stated limit rather than a preference.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
